

AI-Based Air Pollution Detection & Monitoring System

Prepared by: Senthilmurugan – B.E. ECE, Project: AI Immersion

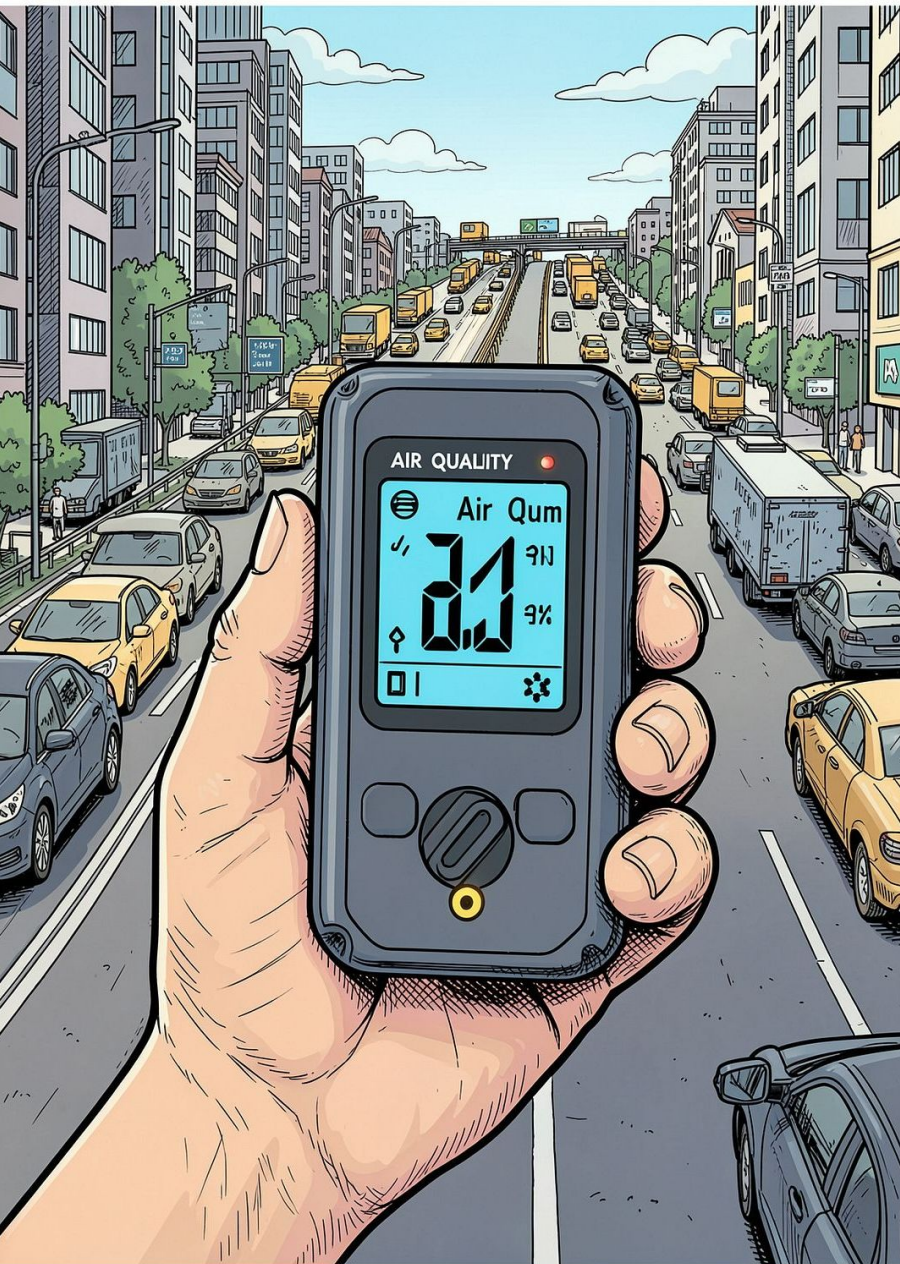
25107152



Introduction

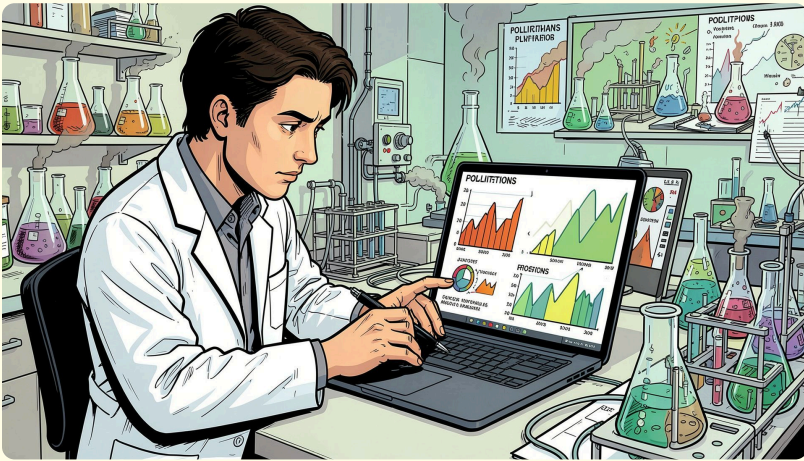
Air pollution is the presence of harmful substances in the atmosphere from vehicles, industry, construction, and waste burning. Rising pollution harms human health, wildlife, and ecosystems. AI enables continuous, scalable monitoring and rapid analysis of air-quality data to inform decisions and alerts.





Problem Statement

Pollution varies by time and place. Manual monitoring is time-consuming and sparse, leaving people unaware when air becomes unhealthy. We need a smart, automated system to continuously detect and classify pollution and issue timely, localized warnings.



Project Objectives

- Continuously monitor air-quality conditions across locations.
- Identify and quantify major pollutants (PM2.5, PM10, CO, NO₂, SO₂).
- Analyze features and train AI/ML models for classification.
- Classify air quality into actionable levels (Good → Very Poor).
- Present clear, user-friendly information and alerts.

Major Air Pollutants



PM_{2.5}

Fine particles from combustion and smoke. Penetrate deep into lungs; linked to cardiovascular and respiratory illness.



PM₁₀

Coarse dust from construction and roads. Causes throat and lung irritation and reduced visibility.



CO

Carbon monoxide from incomplete combustion. High exposure reduces oxygen delivery in the body.



NO₂

From vehicle and industrial emissions. Irritates airways and contributes to ozone formation.

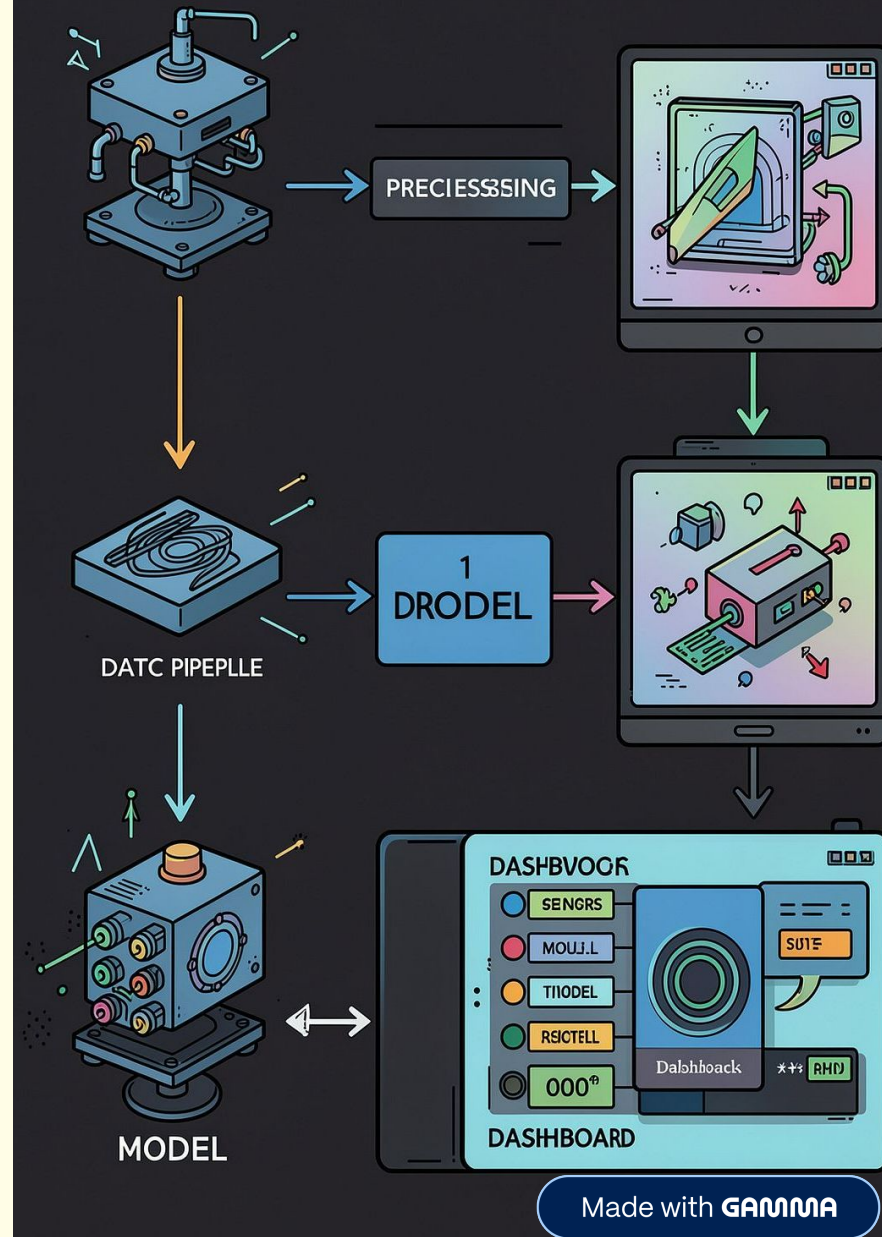


SO₂

From burning sulfur-containing fuels and certain industries. Causes respiratory irritation and acid rain precursors.

Proposed System Overview

We combine sensor or dataset inputs with AI to classify local air quality in real time. Workflow: Air-quality data → Preprocessing → Feature analysis → AI/ML model → Pollution classification → Dashboard & alerts. Categories: ● Good – ● Moderate – ● Poor – ● Very Poor.



AI Methodology – Steps



- **Data Collection:** Gather sensor readings or curated datasets of PM2.5, PM10, CO, NO₂, SO₂, meteorological data.
- **Data Preprocessing:** Clean missing values, calibrate sensors, normalize and align timestamps.
- **Feature Analysis:** Select relevant features, compute moving averages, lag features, and meteorological interactions.
- **Model Training:** Train classification models (e.g., Random Forest, SVM, or lightweight neural nets) with cross-validation.
- **Prediction & Alerts:** Produce per-location air-quality labels and trigger user-facing alerts.

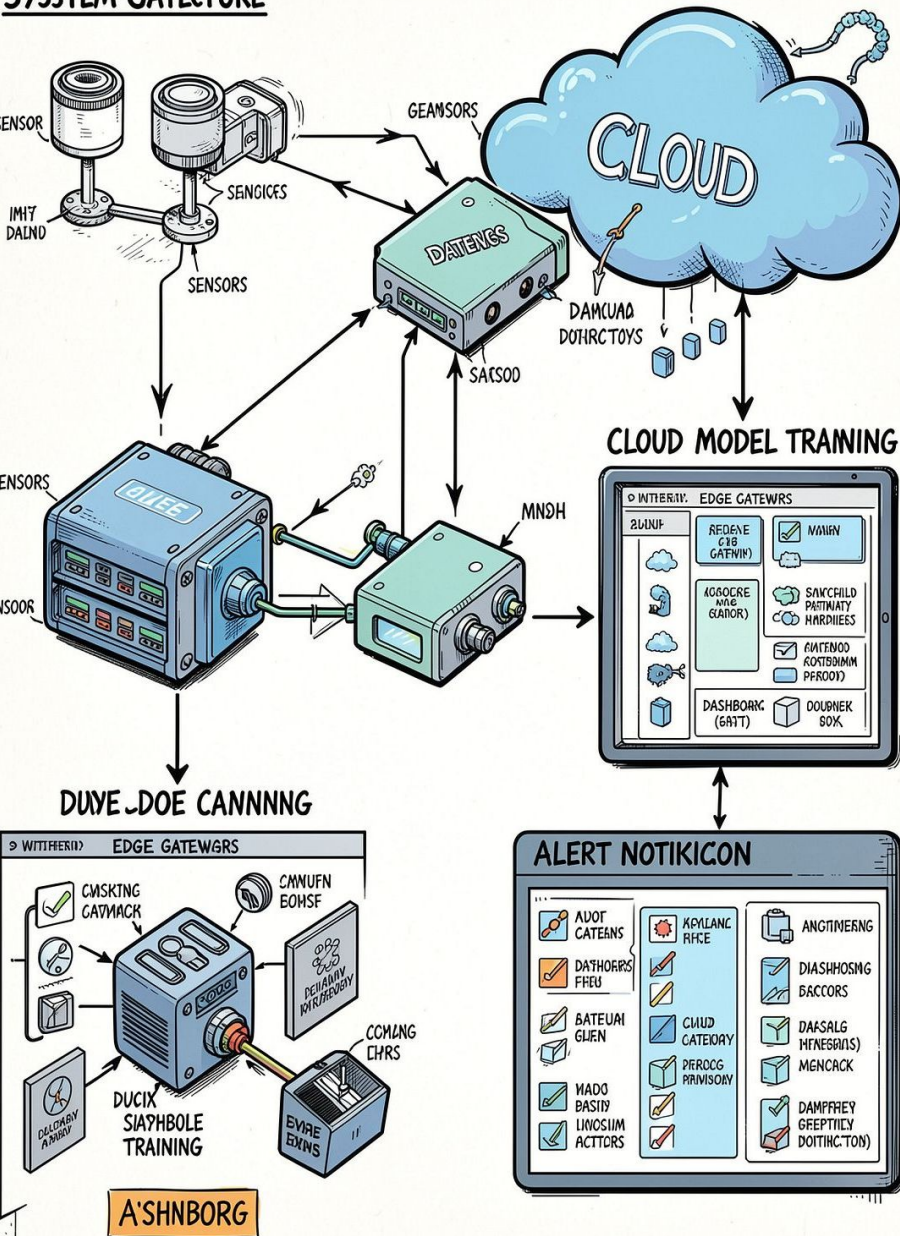


Technologies & Tools

Key tools selected for implementation:

- **Python:** Core language for data processing and modeling.
- **Pandas / NumPy:** Data handling and numerical operations.
- **Scikit-learn:** Classical ML models and evaluation.
- **Streamlit / Web UI:** Lightweight dashboard & visualization.
- **Air Quality Dataset / IoT Sensors:** Training, validation, and live data feed.

SYSTEM GATECTURE



System Architecture

Input: sensors or dataset → Preprocessing: cleaning & calibration → Feature extraction: pollutant and meteorology features → AI/ML Model: trained classifier → Prediction: pollution level → Output: dashboard, push alerts, APIs for integrations. Designed for modular IoT deployment and cloud-assisted model updates.

Advantages, Limitations & Future Work



Smart Cities

Spatially map air quality to guide policy and public health measures.



Industrial Monitoring

Track emissions near factories to ensure compliance and detect incidents.



Traffic Monitoring

Identify hotspots where vehicle emissions degrade local air quality.



Public Alerts & Apps

Deliver location-based warnings and guidance to vulnerable users.

Advantages

- Automates continuous monitoring and fast classification.
- Scales to many locations and processes large datasets.
- Easy-to-interpret outputs and integration with IoT.

Limitations

- Model accuracy depends on dataset quality and sensor calibration.
- Local weather, microclimates, and sensor drift complicate predictions.

Future Scope

- Integrate real-time IoT networks and mobile apps for localized alerts.
- Evaluate advanced deep-learning models and transfer learning.
- Embed adaptive calibration and weather-aware forecasting.